Proposal: Reconstruction of Undersampled MRI Data to Reduce Scan Time

Rough Draft

Student: Shreeya Maharjan
Course: ECE 3090 — Engineering Project Management
Project Adviser: Dr. Gibbons
Weber State University
Project Duration: Fall 2026 – Spring 2027

Executive Summary

MRI scans are essential for diagnosing many medical conditions, but they are slow. Long scan times increase cost, reduce patient access, and cause discomfort. This project proposes implementing and evaluating computational methods that reconstruct high-quality MRI images from deliberately undersampled data, reducing the time required to complete a scan.

MRI scanners collect data in a mathematical space called k-space. If only a fraction of k-space lines are collected, the scan finishes faster. However, a standard image reconstruction then produces artifacts and blurring. This project investigates signal-processing techniques—including compressed sensing and iterative regularization—that can recover a clean image from incomplete measurements. Quality will be measured against fully sampled ground-truth images using SSIM and PSNR metrics, with a target SSIM above 0.85 at four-times acceleration.

A unique aspect of this project is access to real raw k-space MRI data provided by the University of Utah, allowing validation on genuine clinical acquisitions. All software will be written in Python using open-source libraries, keeping costs under $75.

Commercial products such as Siemens Deep Resolve and GE AIR Recon DL already apply deep learning to accelerate MRI, but they are proprietary and tied to specific hardware systems. Published research by Rastogi et al. (*Lancet Oncology*, 2024) confirms that neural-network reconstruction at ten-times acceleration can meet clinical quality standards. This project builds on that work using open tools and real data, contributing academic and research value.

I have worked with signal processing, embedded systems, and Python throughout my coursework and projects. I have already completed Signals and Systems, Digital Circuits, and Microelectronics, and will finish Digital Signal Processing and Microprocessor Architecture before the project begins. I will also be taking Image Processing in Fall 2026, which will directly support the reconstruction and evaluation work in this project.

Technical Description

Project Overview

The objective of this project is to implement and evaluate a software pipeline that reconstructs MRI images from undersampled k-space data. The system will simulate accelerated MRI acquisition by applying undersampling masks to raw k-space data and then recovering the image using signal-processing reconstruction algorithms. Results will be compared against fully sampled ground-truth images to assess quality.

The system will consist of the following main components:

- Raw k-space MRI dataset provided by the University of Utah
- Undersampling masks at accelerated factor
- Baseline zero-filled inverse Fourier transform reconstruction
- Regularized reconstruction algorithm (e.g., compressed sensing or iterative method)

- Image quality evaluation pipeline using SSIM and PSNR metrics
- Segmentation-based Dice score validation on reconstructed images

Statement of Work

The project will involve the following tasks:

1. Obtain and verify raw k-space MRI dataset from the University of Utah.
2. Document dataset characteristics including number of subjects, MRI sequence type, and matrix size.
3. Design and implement undersampling masks at acceleration factors R=4 and R=8 using random and center-weighted patterns.
4. Verify that each mask removes the correct fraction of k-space lines to within ±1%.
5. Implement a baseline zero-filled inverse Fourier transform reconstruction as a reference.
6. Implement at least one regularized reconstruction algorithm.
7. Implement SSIM and PSNR evaluation against fully sampled ground-truth images.
8. Conduct a systematic comparison of reconstruction methods across both acceleration factors.
9. Perform a segmentation-based Dice score validation on a subset of reconstructed images.
10. Produce a final written report, documented codebase, and presentation.

Major Project Phases

Phase 1 – Background Research and System Setup (Weeks 1–6)

- Review literature on MRI acceleration and reconstruction methods
- Obtain dataset access and verify data format and integrity
- Set up Python programming environment and project repository
- Identify and install required open-source libraries

Phase 2 – Undersampling and Baseline Reconstruction (Weeks 7–13)

- Implement undersampling masks
- Verify mask accuracy against target acceleration factor
- Implement baseline zero-filled reconstruction
- Build SSIM and PSNR evaluation pipeline

Phase 3 – Algorithm Development and Testing (Weeks 14–22)

- Implement regularized reconstruction algorithm
- Compare methods across both acceleration factors

- Perform Dice score segmentation validation
- Tune algorithm parameters to meet quality targets

Phase 4 – Documentation and Final Preparation (Weeks 23–26)

- Write final technical report
- Clean and document codebase
- Prepare final presentation

A Preliminary Design Review (PDR) will occur after approximately 13 weeks, and a Final Design Review (FDR) will occur after 26 weeks.

Expected Benefits and Limitations

Benefits

- Reduces MRI scan time, increasing patient throughput and lowering cost per scan
- Improves patient comfort by shortening time spent in the scanner
- Reduces motion artifacts caused by long scan durations
- Produces an open-source, hardware-independent reconstruction pipeline
- Contributes to research aimed at expanding MRI access in limited-resource settings

Limitations

- This is a software simulation study and does not involve MRI scanner hardware
- Reconstruction times on a personal computer will be longer than clinical real-time systems
- The pipeline will be validated on a specific dataset and may require tuning to generalize to other scanners

Estimated Project Costs

Labor Estimate

Task	Estimated Hours
Background study and literature review	40
Dataset acquisition and preprocessing	30
Algorithm implementation and debugging	100
Testing, evaluation, and optimization	60
Report writing and documentation	50
Presentation preparation	20
Total	**300 hours**

Materials and Equipment

Item	Estimated Cost
Python and open-source libraries (NumPy, SciPy, PyTorch)	$0
University of Utah k-space dataset (provided)	$0
Cloud computing, optional (e.g., Google Colab Pro)	$50–$70
Printing for final report submission	$5
Total	**$55–$75**

Total Project Cost

Estimated labor and materials combined: approximately $55–$75 in expenses plus 300 hours of engineering work. No specialized hardware purchases are required.

Funding Sources

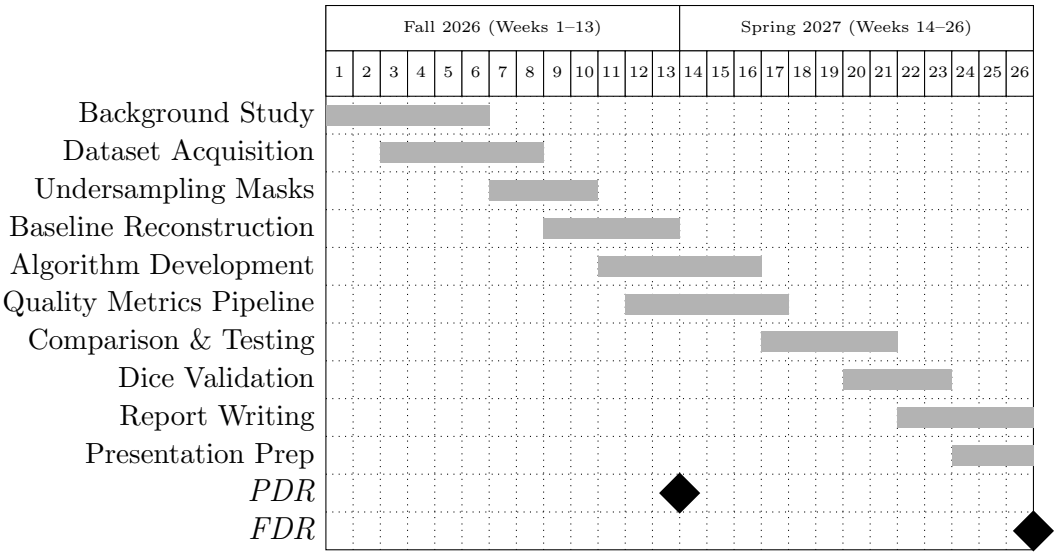
Potential funding sources include:

- Weber State University ECE Department internal grants for senior design projects
- Undergraduate research funding available through the department
- Personal funding by the student for minor expenses such as printing

Project Timeline

Phase	Duration
Background research and environment setup	Weeks 1–4
Dataset acquisition and verification	Weeks 3–6
Undersampling mask implementation	Weeks 7–10
Baseline reconstruction and quality metrics	Weeks 9–13
Preliminary Design Review (PDR)	Week 13
Regularized algorithm development	Weeks 11–16
Algorithm comparison and testing	Weeks 17–21
Dice score validation	Weeks 20–23
Report writing and documentation	Weeks 22–26
Presentation preparation	Weeks 24–26
Final Design Review (FDR)	Week 26

Gantt Chart



Diamonds indicate the Preliminary Design Review (PDR) at week 13 and the Final Design Review (FDR) at week 26.

Deliverables

At the completion of the project, the following deliverables will be provided:

- Fully documented Python codebase including undersampling, reconstruction, and evaluation scripts
- Dataset documentation describing preprocessing steps and any data use requirements
- Preliminary Design Review presentation and written summary
- Final technical report describing methodology, results, and conclusions
- Final Design Review presentation with demonstrated reconstruction pipeline
- Tables and figures summarizing SSIM, PSNR, and Dice scores for all tested configurations

Qualifications

Computer Engineering student at Weber State University with experience in:

- Signal processing and Fourier-domain analysis
- Embedded systems and microcontrollers
- Scientific computing and programming in Python and C
- Image reconstruction and data analysis

Relevant coursework includes upper-division engineering courses focused on signals and systems, digital signal processing an Image Processing in Fall 2026, will also be taking which will directly support the reconstruction and evaluation work in this project.

Approvals

Signatures below indicate approval of this project proposal. Do not sign until the proposal has been graded and revised if necessary.

Student: Shreeya Maharjan

Signature: _____ Date: _____

Project Adviser: Dr. Gibbons

Signature: _____ Date: _____

ECE Faculty:

Signature: _____ Date: _____

References

1. A. Rastogi et al., "Deep-learning-based reconstruction of undersampled MRI to reduce scan times: a multicentre, retrospective, cohort study," *Lancet Oncology*, vol. 25, pp. 400–410, Mar. 2024.
2. A. Razumov, O. Rogov, and D. V. Dylov, "Optimal MRI undersampling patterns for ultimate benefit of medical vision tasks," *Magnetic Resonance Imaging*, vol. 103, pp. 37–47, 2023.
3. D. Huo and D. L. Wilson, "Robust GRAPPA reconstruction," in *Proc. IEEE International Symposium on Biomedical Imaging (ISBI)*, 2006, pp. 37–40.

Shreeya Maharjan

github.com/shreeyamhr linkedin.com/in/shreeyamhr
shreeyamaharjan@mail.weber.edu (571) 602-3557
Ogden, UT

Detail-oriented Computer Engineering student at Weber State University with good GPA and strong foundations in embedded systems, software and hardware programming, signal processing, circuit design, and computer architecture. Prepared to contribute hands-on design, analysis, and troubleshooting skills.

Education

Bachelor of Science in Computer Engineering, *Weber State University* expected April 2027
- GPA: 3.6

Assoc. Pre Engineering, *Weber State University* April 2025
Relevant coursework: Data Structures Algorithms, Computer Architecture, Digital Design, Signal Processing, Embedded System

Skills

Programming Languages
C/C++ | Python | Assembly | SQL | VHDL

Software & Design Tools
Linux | Git/GitHub | FPGA (Basys3) | Microcontrollers | Vivado | KiCAD | Ltspice

Test and Measurement tools
Oscilloscope | Multimeter | Function Generator | Power Supply | Signal Analysis

Circuit and system Knowledge
Analog and digital circuit analysis | Power supply fundamentals and specifications | Electrical troubleshooting and diagnostics | Microprocessor and embedded system control | Signal processing and filter design

Professional Skills
Excellent Interpersonal communication | Code documentation | Project management

Technical Experience

IT Technician **May 2025 — Present**
Student Affair Technology *Weber State University, UT*
- Managed system imaging, deployment, and recovery workflows supporting large-scale lab infrastructure across multiple facilities
- Diagnosed and resolved hardware, software, and network issues across Windows environments, maintaining consistent uptime for hundreds of users.

Computer Lab Team Lead **Feb 2023 — May 2025**
Student Affairs Technology *Weber State University, UT*
- Led and trained a team of 8+ lab assistants, standardizing procedures across multiple labs to improve operational consistency.
- Served as primary technical resource for hardware and software issues across a high-traffic, multi-platform environment.

Projects

Embedded Systems Project — 2 Player LED Ping-Pong Game
- Designed and implemented a real-time embedded system using 8051 assembly with interrupt-driven architecture.
- Developed timing-critical control logic for responsive input handling and dynamic LED output.
- Optimized system performance by minimizing latency in signal updates and improving execution efficiency.

Microprocessor Controller Development
- Designed a finite state machine (FSM) to control microprocessor instruction sequencing and execution.
- Integrated control logic into processor architecture to enable efficient and autonomous operation.

Calculator Datapath and ALU Development
- Designed and integrated an 8-bit ALU into a calculator datapath using synthesizable VHDL.
- Applied sequential and combinational RLT design principles to create a hierarchical component structure.
- Developed an automated VHDL testbench to verify RTL functionality, reducing manual testing and improving verification accuracy.

Awards

East Dean's Scholarship 2025-Present
Louis F. Moench Scholarship 2023-Present